

UNIT I

BUILDING MATERIALS

➤ STONES

➤ BRICKS

➤ CEMENT

➤ CONCRETE

➤ STEEL



CONCRETE

- Concrete is the most versatile material for all types of construction works and has been used for innumerable construction works, either as plain concrete or as reinforced cement concrete or as precast concrete, or prestressed concrete or in many other forms.
- The various constituents of concrete are cement, water, fine aggregate, and coarse, aggregates. In Reinforced Cement Concrete Steel is also used.
- Cement: Properties and various types of cement have already been discussed previously.

Role of Aggregates in Concrete

60-80% of the volume of concrete is occupied by aggregates.

Main significance:

- Cost
- Dimensional stability
- Strength and stiffness
- Abrasion resistance



Aggregate: (Coarse and Fine)

- These are the inert or chemically inactive materials which form the bulk of cement concrete.
- These aggregates are bound together by means of cement. The aggregates are classified into two categories, Fine and coarse.
- The material which is passed through 4.75mm size sieve is termed as **fine aggregate**. Usually natural river sand, issued as a fine aggregate.
- The material which is retained on 4.75 mm size sieve termed as a coarse aggregate. Broken stone is generally used as a **coarse aggregate**.

Water:

- Water which is used for making concrete should be clean and free from harmful impurities such as oil, alkali, acid etc.
- In general water which is fit for drinking should be used for making concrete.

Grades of Concrete

- Concrete as per IS456:2000 is classified into three groups as ordinary concrete, Standard concrete and High strength concrete.
- M10, M15 and M20 are ordinary concrete, M25, M30, M35, M40, M45, M50 and M55 are grouped as Standard concrete and M60, M70, M75 and M80 are grouped under High strength concrete.
- The letter 'M' refers to the mix and the number indicates the specified compressive strength of that mix at 28 days expressed in or Mega Pascal (MPa) or N/mm^2 .
- For lean concrete bases and simple foundations for masonry walls M5 and M7.5 grades of concrete may be used. These mixes need not be designed

Grades of Concrete contd.....

- Grades of concrete lesser than M20 are not to be used in RCC work.
- For general guidance, the nominal mix proportions (volume ratio) correspond approximately to the different grades as follows:
 - ❖ M 5 - 1:5:10 (Cement : Fine aggregate : Coarse aggregate)
 - ❖ M 7.5- 1:4:8
 - ❖ M 10 - 1:3:6
 - ❖ M 15 - 1:2:4
 - ❖ M 20 - 1:1.5:3
 - ❖ M 25 - 1:1:2

Gain of strength with age contd.....

- The concrete develops strength with continued hydration.
- The rate of gain of strength is faster to start with and the rate gets reduced with age.
- It is customary to assume the 28 days strength as the full strength of concrete. Actually concrete develops strength beyond 28 days also. The variation of strength with age is given below

Age of curing (Days)

3

7

28

Approx. Strength achieved

$1/3^{\text{rd}}$ of target strength

$2/3^{\text{rd}}$

90 %

Strength of Concrete

- Strength of concrete is its resistance to rupture.
- It may be measured in number of ways, such as strength in compression, in tension, in shear or in flexure.
- The compressive strength of concrete is generally determined by testing cubes or cylinders made in laboratory or field. The size of the mould should be 150 mm x 150 mm x 150 mm.
- Based on the compressive strength, only the concrete is graded.

Strength of Concrete contd.....

The strength of the concrete is mainly depend on the following factors:

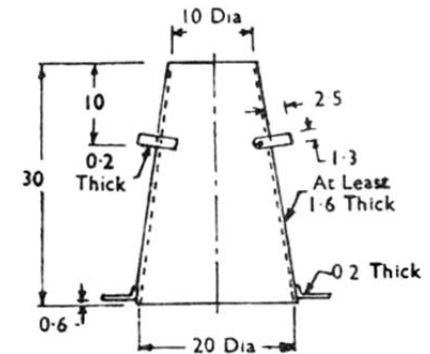
- Quality of materials and grading of the aggregates
- Water cement ratio
- Cement content
- Age of concrete; and
- Methods of mixing, placing, compacting and curing.

Workability of concrete

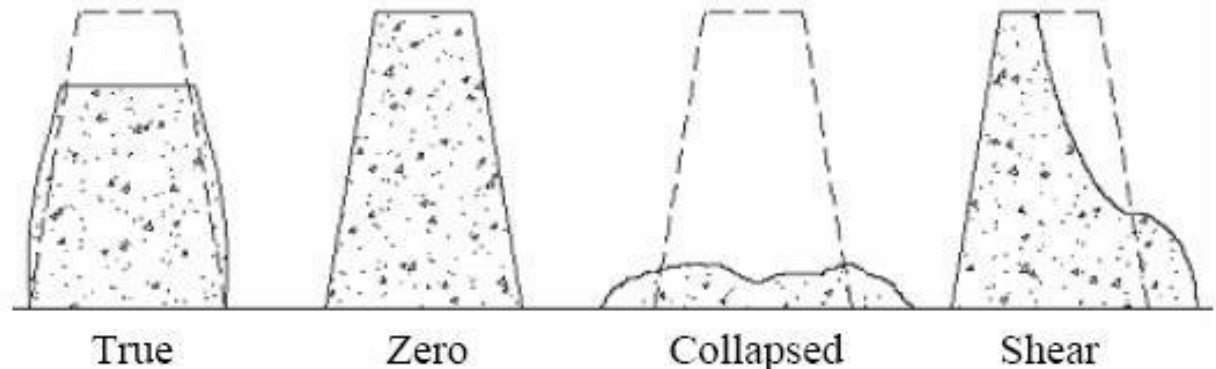
- The workability of concrete indicates the ease with which it can be mixed, placed and compacted.

Slump Test to define workability

- Slump test is the most commonly used method of measuring workability of concrete which can be employed either in the laboratory or at site of work.



All dimensions in Centimeters

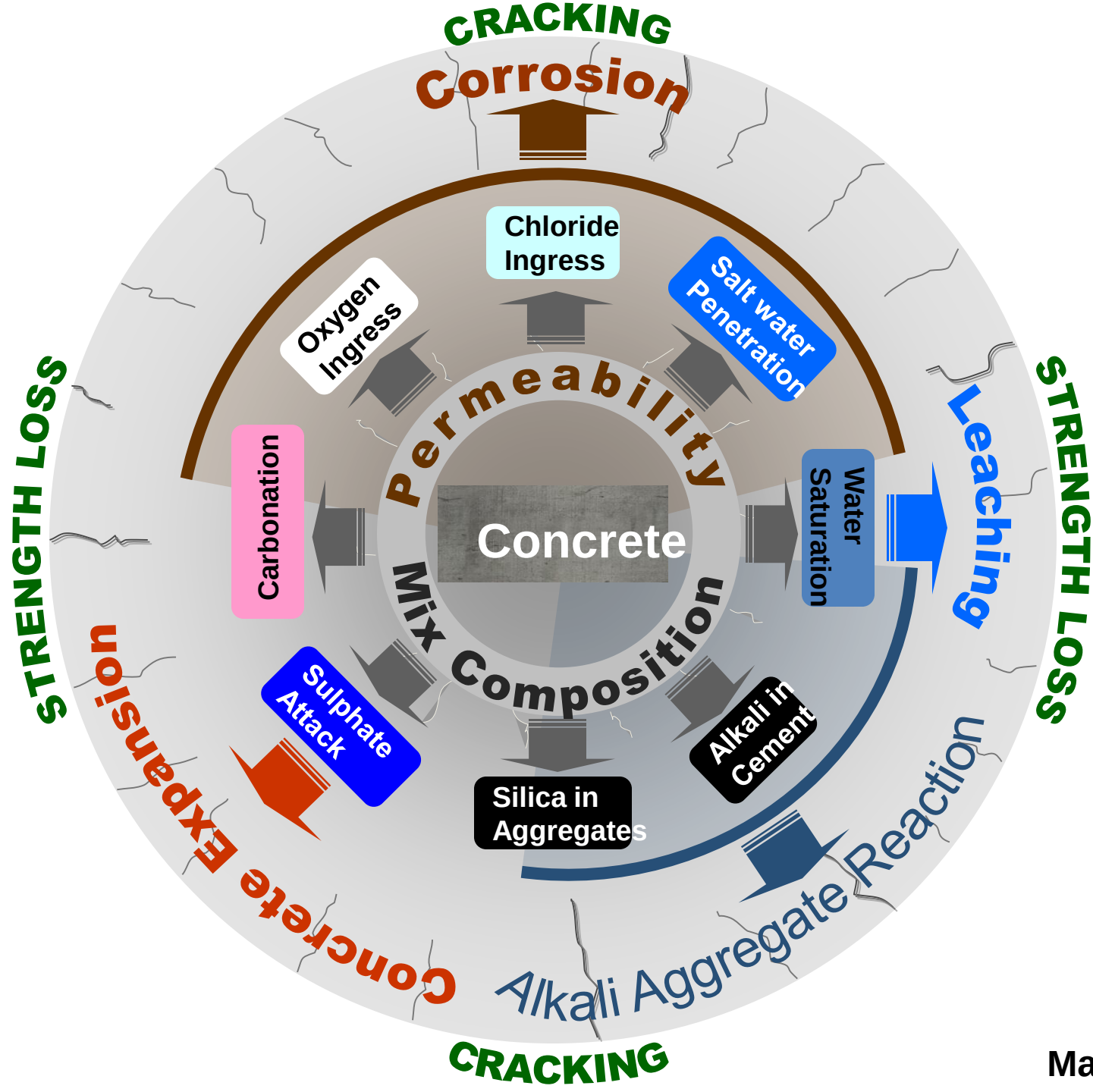


Durability of Concrete

- The concrete possesses a high durability value, it is not much affected by atmospheric actions.
- On the contrary, with the age, the concrete goes on hardening, there by increasing in strength.

Water Cement Ratio

- The strength of concrete depends upon the quantity and quality of its ingredients i.e. cement, aggregate and water.
- General assumption is that strength of concrete directly depends up on the quantity of cement.
- If cement is more, strength will be more but this assumption is not correct because strength of concrete also depends upon water cement ration.



Operations in Concreting

The following processes are involved in making concrete to achieve desired strength and durability.

1. Storing of material (Cement, aggregate)
2. Batching of material (By volume, by weight)
3. Mixing of concrete (By hand, by machine)
4. Transportation of concrete.
5. Placing of concrete.
6. Curing of concrete



Operations in Concreting contd.....

1. Storing of Material

Cement:

- It is a fine powder and also hygroscopic in nature i.e. it absorbs moisture from air or free water and starts setting.
- Hence the warehouses constructed for its storage must fulfill the basic requirements.
- Cement stored for long time should be checked before its use.

Aggregates:

- It is essential that aggregate should be free from deleterious materials, organic matters such as tree leaves, vegetable wastes, animal refuse etc.
- It should have uniform moisture content and proper grading of aggregates.

Operations in Concreting contd.....

2. Batching of Materials

- Batching means measurement of ingredients of concrete for proper mixing.
- Normally such a quantity is mixed in one batch, which can be transported, placed and compacted within time i.e. before initial set takes place.
- ❖ Batching is of two types.
 - Volume Batching.
 - Weight Batching.

2. Batching of Materials contd.....

Measurement of Cement:

- Cement is always measured by weight.
- A batch of concrete should always consume full number of bags. For this purpose weight, of cement bag is taken as 50 kg.

Measurement of Water:

- Water is generally measured by volume because its weight can not be easily calculated.

Measurement of Aggregate by Volume:

- For these purpose generally wooden boxes of capacity equivalent or part of one cement bag i.e. 35 liters are used. These boxes are known as Petties or Farmas or Gauge Box.

2. Batching of Materials contd.....

Weigh Batching:

- As per the name, ingredients of concrete are measured by weight.
- This is more accurate and is generally used where high quality concrete is required.
- As described earlier generally one cement bag or its multiple is the basic unit.
- In this system allowance for water present in aggregate is made, but bulking has no effect.

Operations in Concreting contd.....

3. Mixing of Concrete:

- i) Hand Mixing
- ii) Machine Mixing.

i) Hand Mixing: It is adopted for small works where quantity of concrete required is small. In it various ingredients of concrete are mixed by hand.

ii) Machine Mixing: For producing large quantity of concrete at a faster rate and at lesser cost, machine mixing is adopted.

- A. Continuous mixer
- B. Batch type mixer.

3. Mixing of Concrete contd.....

A. Continuous mixer:

- It mixes and delivers concrete just as a steady stream of concrete, till it is in operation.
- This type of mixers is not so common in use in India.

B. Batch type mixer:

- Mixes and discharge each loads of material separately. These are of two types.
 - a. Tilting mixers.
 - b. Non-Tilting or Rotary Mixers.

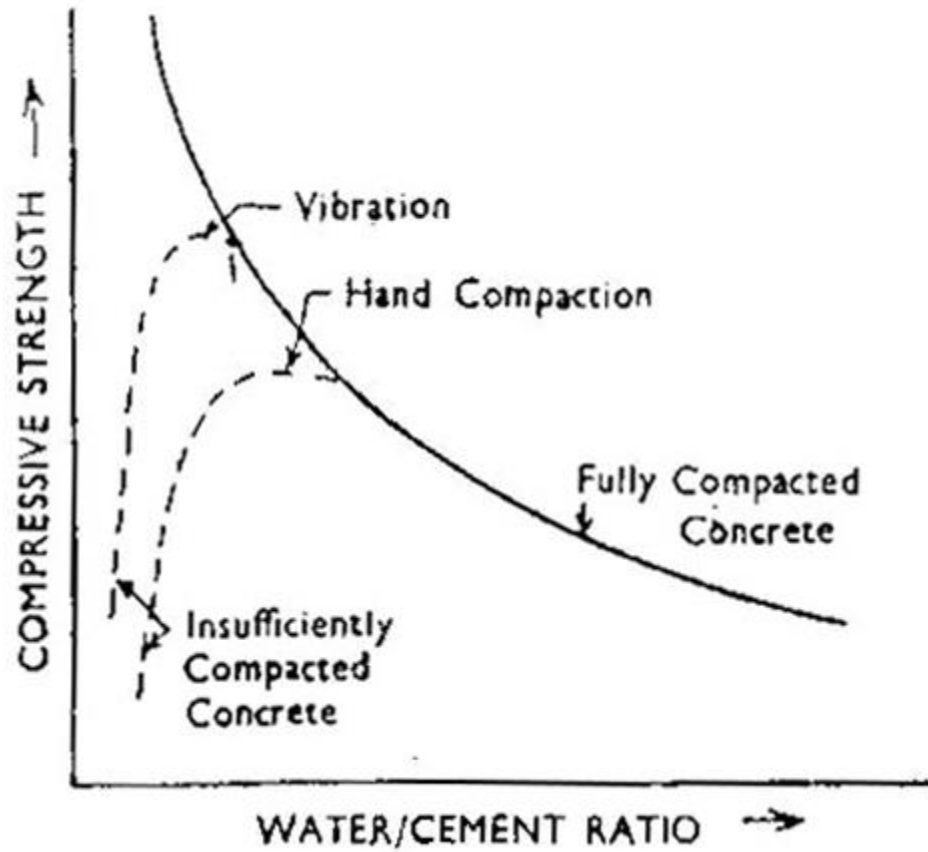
Rotating mixer



Tilting mixer



Manual & machine compaction



The relation between strength and water/ cement ratio of concrete

Operations in Concreting contd.....

4. Transportation of concrete:

- As the initial setting time of cement is generally 30 minutes, hence mixing, transportation, placing and compaction should be completed within this time.
- In no case this time should not exceed one hour after initial setting time.

5. Placing of Concrete:

- As far as possible concrete should be placed in single thickness.
- In case of deep sections, concrete should be placed in successive horizontal layers and proper care should be taken to develop enough bond between successive layers.

Placing of concrete



Operations in Concreting contd.....

6. Curing of Concrete:

- Concrete surfaces are kept wet for a certain period after placing the concrete.
- The period of curing depends on the type of cement and nature of work.
- For ordinary Portland cement, the curing period is 7 to 14 days. If rapid hardening cement is used, curing period can be considerably reduced.
- It can be done by spraying and ponding of water or covering the concrete with moist earth, sand, or wet gunny bags.

Types of Concrete

- Plain Cement Concrete
- Reinforced Cement Concrete
- Pre Stressed Concrete
- Pre-Cast Concrete
- High Strength or High Performance Concrete
- Self Compacting or Self Consolidating Concrete
- Fibre Reinforced Concrete
- Light weight / Foamed Concrete
- Shotcrete
- Porous / pervious / permeable Concrete
- High Density Concrete
- Heat Resistant and Refractory Concrete
- Underwater Concrete
- Recycled Aggregate Concrete
- Geopolymer Concrete

1. Plain Cement Concrete

- It is a mixture of cement, sand pebbles or crushed rock and water. It possesses the following important properties.
 - It is free from corrosion
 - It has a high compressive strength
 - It binds rapidly with steel
 - It has a tendency to be porous.

2. Reinforced Cement Concrete

- Plain cement concrete is strong in compression but weak in tension.
- To make this efficiency better steel bars known as reinforcement are embedded in concrete.



Plain concrete

Reinforced concrete



High Strength / High Performance Concrete

- High performance concrete is that which satisfies the requisites in terms of all the critical aspects of the fabrication and utilization, at the lowest possible cost.
- There is no unique high performance concrete. The mix design depends on the application.
- High performance concrete refers to a type of concrete that offers more than just high strength.
- Generally, high performance implies both high compressive strength and durability, today.

High Strength / High Performance Concrete

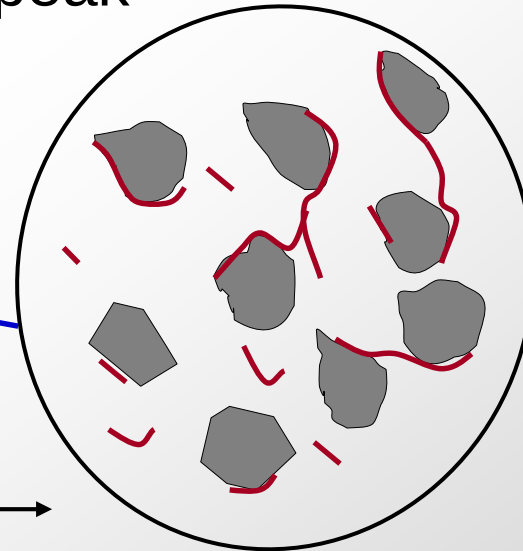
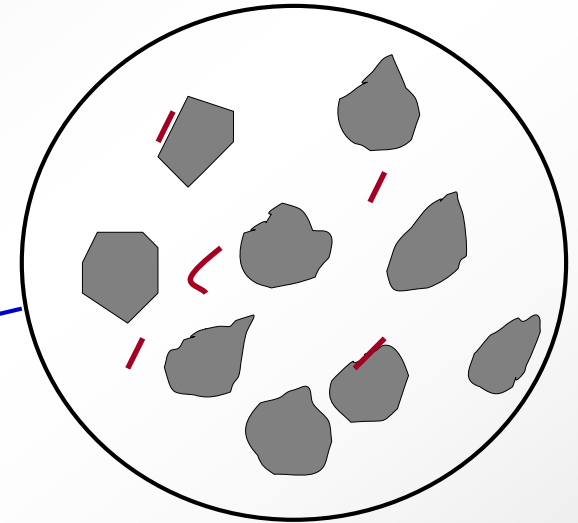
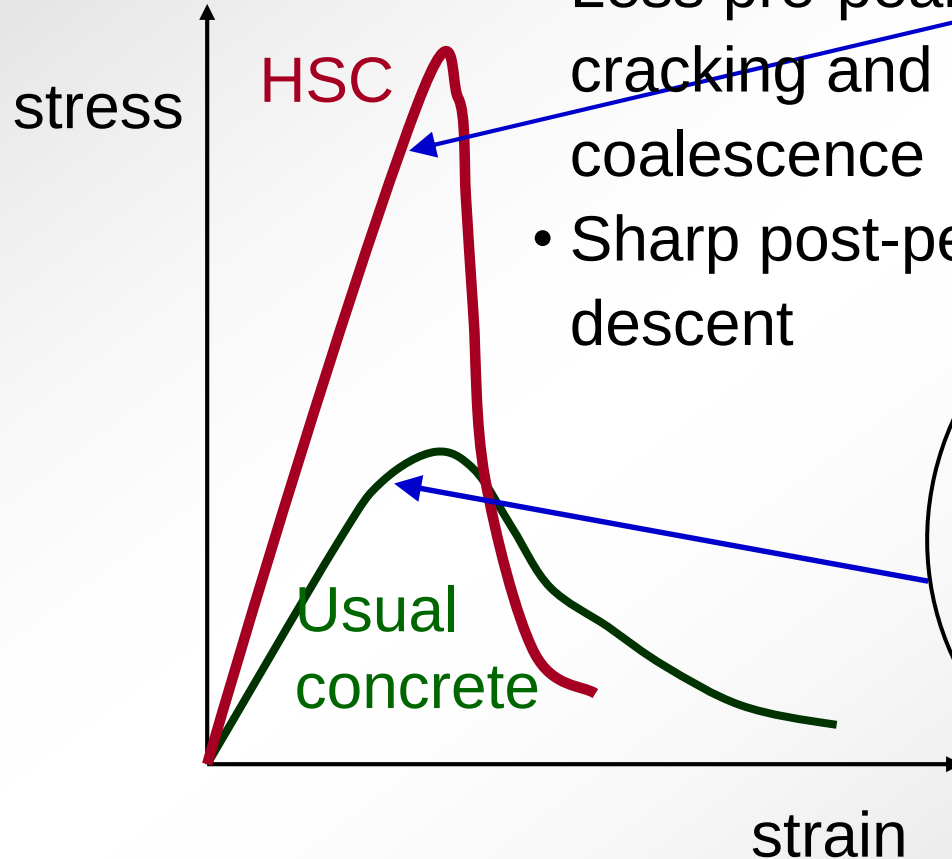
Important Characteristics

- Good quality material components
- Dense packing of aggregates
- Low water/cement ratio, normally less than 0.35
- Incorporation of one or more pozzolanic mineral admixtures, such as silica fume, fly ash, slag & metakaolin
- Adequate type and dosage of superplasticizer to obtain the desired workability
- Proper curing to avoid early cracking due to shrinkage and thermal stresses

High Strength / High Performance Concrete

Mechanical Behaviour

- More pre-peak linearity
- Less pre-peak cracking and no crack coalescence
- Sharp post-peak descent



High Strength / High Performance Concrete

- Example of Applications

Hibernia Offshore Platform, 1997



- More than 110 m high
- Designed to resist iceberg impact
- Vertical parts were slipformed; placing rate was 1 m/day

Most concrete had unit weight of 2200-2250 kg/m³, elastic modulus > 32 GPa, characteristic compressive strength = 69 MPa

Self-Compacting Concrete

- It is a concrete that is capable of flowing within a mould or formwork, filling it completely, passing through the reinforcement and consolidating under its own weight.
- Important characteristics in the fresh state:
 - Flowability or filling ability
 - Passing ability
 - Stability or resistance against segregation

The mechanisms that govern these properties are the fluidity and internal cohesion of the fresh concrete.



Self-Compacting Concrete

General Characteristics

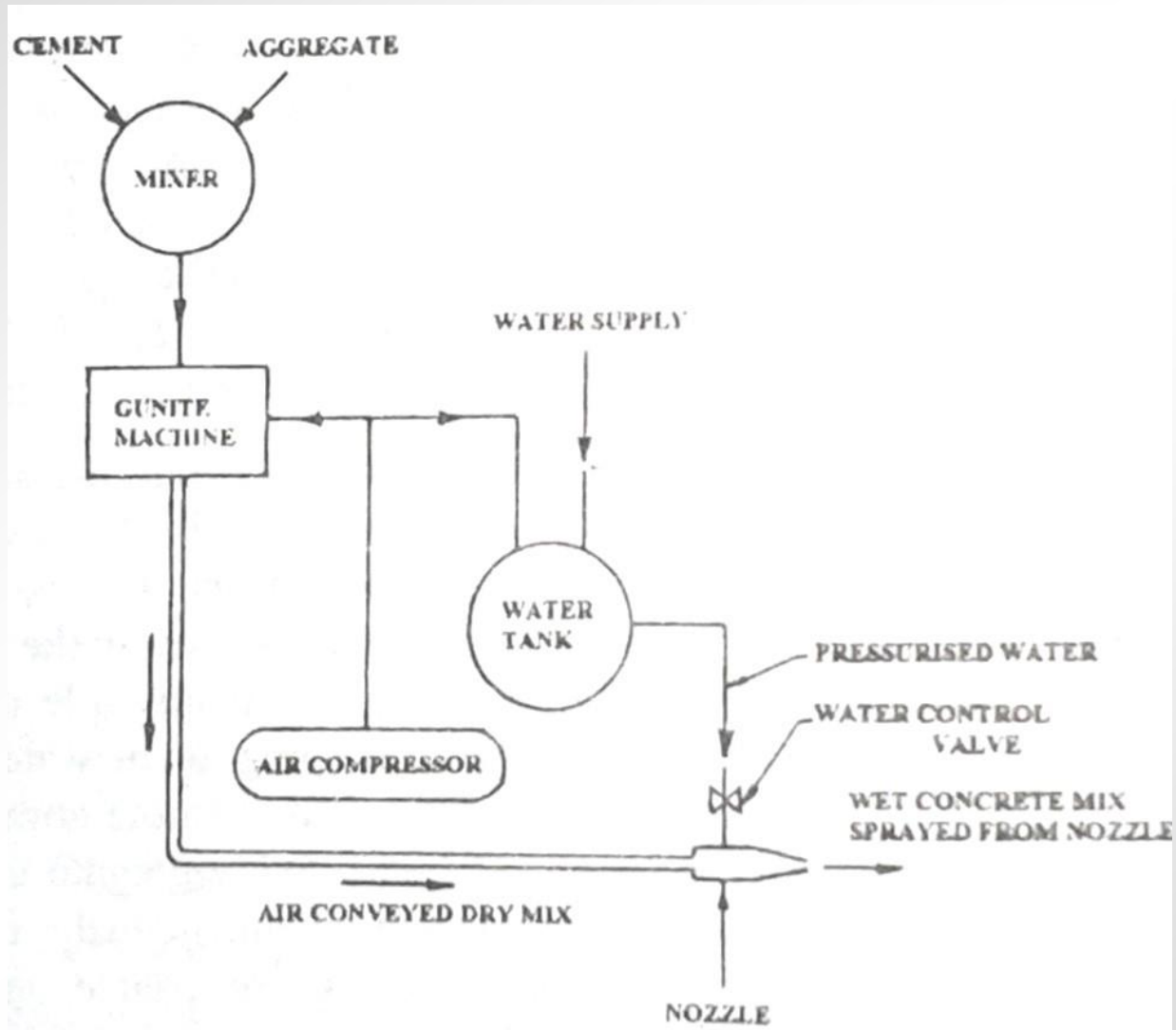
- Maximum aggregate size limited to 25 mm
- Paste content of about 400 litre/m³
- Water/fines ratio of 0.31-0.36
- Fines content of 500-600 kg/m³ (i.e., cement, mineral admixtures, fillers)
- Coarse aggregate/sand ratio of about 50/50
- Superplasticizer for fluidity
- Viscosity modifier for stability and robustness

Shotcrete / Sprayed Concrete

- Concrete or mortar that is placed and compacted by projecting it pneumatically at high velocity onto a surface.
- Wet-mix shotcrete: All major ingredients are mixed before introduction in the hose. Compressed air is introduced to the material at the nozzle. If an accelerator is used, it is normally added at the nozzle.
- Used in tunnel lining, slope stabilization and repair.
- Possibility of use in sloped and shell roof construction

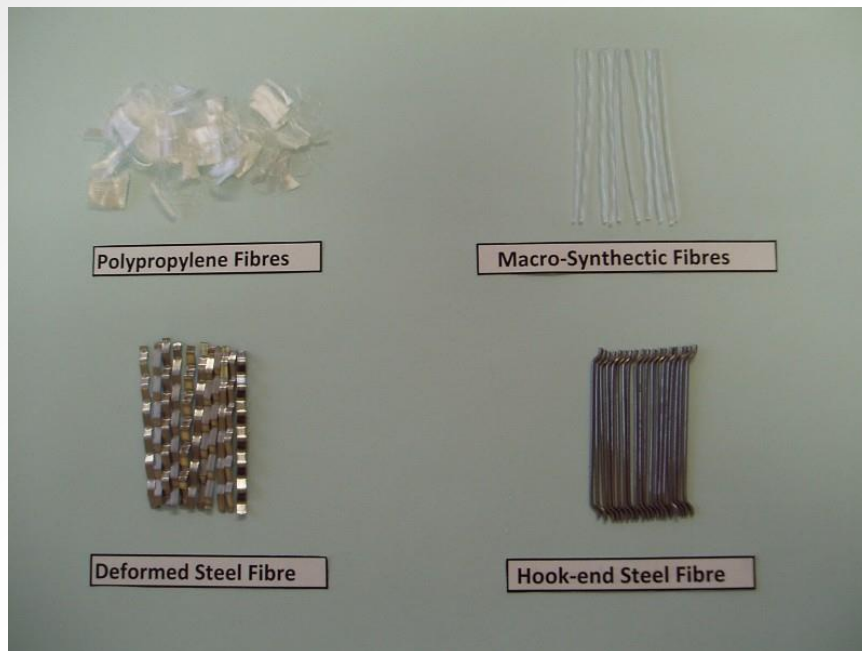


Shorcrete mixing process



Fibre Reinforced Concrete (FRC)

- Conventional fibre performance concrete is that which has a homogeneous distribution of randomly-oriented short fibres.
- Fibres normally used are steel or polypropylene. Steel fibres improve the properties of the hardened concrete (i.e., increase the post-cracking strength). Asbestos, glass, nylon or coconut fibers are also used.



Fiber Reinforced Concrete (FRC)

- The fibres are generally:
 - much shorter than the dimensions of the concrete element
 - stronger and can elongate more than the matrix under tension
 - introduced in the matrix during the mixing of the concrete
- The advantages are:
 - It has thin sections
 - More durable
 - Increases toughness of concrete
 - Decrease plastic shrinkage cracking



Lightweight Concrete (LWC)

- Lightweight concrete has a maximum density of 1900 kg/m^3 and is achieved by using low-density aggregates.
- Intermediate-density concretes, where part or all the normal-density coarse aggregates are replaced with structural-grade low-density aggregates, have densities of $1900\text{-}2200 \text{ kg/m}^3$.
- Low density aggregates used could be natural (expanded by heating or of volcanic origin), such as shale, clay or slate; processed, such as sintered or cold-bonded fly ash.

High Density Concrete (HDC)

- High-density or heavyweight concrete has a density greater than 2800 kg/m^3 , and a typical range of $3200 - 4200 \text{ kg/m}^3$. It is made with aggregates having high density, such as iron ores or pieces of steel (i.e., punchings, shots, rods).
- Densities of up to $3600\text{-}4000 \text{ kg/m}^3$ can be obtained with naturally occurring aggregates (both fine and coarse), such as haematite, magnetite, limonite and ilmenite. For higher densities, steel pieces have to be used.
- The mixes are generally harsh and have a tendency to segregate. Other than this, the mix proportioning is similar to that of normal concrete.

High-Density Concrete

- The main application is in nuclear power plants and proton beam therapy clinics for radiation shielding as the high density facilitates the attenuation of gamma and fast neutron radiation. For the attenuation of moderate and slow neutrons, aggregates containing hydrogen are desirable and so boron-containing aggregates are useful.
- In India, HDC has been used in the nuclear power plants in Rawatbhata and Kalpakkam for radiation shielding.
- Other applications include counterweights and in underwater pipeline coverings.

Heat Resistant / Refractory Concrete

- Heat-resistant concrete can be exposed to temperatures up to 1000 °C, and refractory concrete up to 1900 °C.
- These concretes are made with high-alumina cement, which contains about 40% of alumina and 40% of lime, and the main cementitious compounds are CA and $C_{12}A_7$, whose hydration results in C-A-H.
- Aggregates that can resist high temperatures such as slag need to be used.

Underwater Concrete

- Concrete to be placed under water must be cohesive (to minimise cement washout), flowing (to fill spaces and bond well) and self-compacting (as vibration is not practical).
- Normally, a higher fines content and a viscosity modifying admixture is used in underwater concrete.
- The concrete should be protected such that the fines are not washed away. This can be achieved by using tremies and pumps.

Recycled Aggregate Concrete

- Recycled aggregate concrete generally refers to that made with aggregates derived from construction and demolition waste.
- The majority of such concretes are used in road subbase construction, engineering fill or landfill engineering.
- There is a good potential for its use in building blocks and paver blocks/slabs.

Recycled Aggregate Concrete

Aggregates derived from other industrial wastes:

- **Waste Glass:** Can be used as fine aggregates under some circumstances. Has been used to get a glittery surface in precast concrete. Can be susceptible to alkali-silica reactivity.
- **Granulated or crumb rubber:** Can be used in low-strength flowable concretes. Improves flexibility and thermal insulating properties.
- **Incinerator bottom ash:** Waste from the burning of municipal waste has been used in low strength non-structural concrete.

Geopolymer Concrete

- Geopolymer concrete is one where the binder is made up of fine aluminosilicates activated by alkalis to give poly(sialates). Sialate stands for silicon-oxo-aluminate. Poly(sialates) are inorganic polymers with chain and rings structures of Si^{4+} and Al^{3+} in coordination with oxygen.
- There are a large range of natural materials, such as zeolite, lateritic rock and soils, clay, mica and feldspar, and waste products, such as fly ash and slag, that can be used in the production of geopolymers.
- Potential applications include building and paver blocks, heat- and chemical-resistant composites, and hazardous waste encapsulation.

Pre-stressed Concrete

- In this type of concrete, high tensile steel wires are used as reinforcement instead of mild steel bars.
- ❖ There are two types of prestressing namely
 - i) Pre-tensioning and
 - ii) Post tensioning.

i) Pre-tensioning:

- In pre-tensioning method, the wires are initially stressed and the concrete is cast in the moulds built around the wires.
- The wires released after the concrete attain its strength.

Prestressed concrete



Pre-stressed Concrete contd.....

- The tendency of the wires to return to their original length sets up a compression in concrete, which helps the concrete to resist more tensile stress.

ii) Post-tensioning:

- In post tensioning the wires are placed inside the concrete and then stressed.
- The use of prestressed concrete results in saving of concrete and steel to the extent of 50% and 80% respectively compared to RCC.

No-fines /pervious / permeable Concrete

- It consists of cement, coarse aggregate and water. Thus fine aggregate or sand is eliminated.
- This concrete has been adopted for parking areas, path to swimming pools, etc.,
- The advantages are.
 - It possesses better insulating properties
 - The unit weight of no-fines concrete is about $\frac{2}{3}$ of the unit weight of conventional concrete.
 - There is direct saving in material requirements as the concrete does not require sand.
 - The drying shrinkage is low



Precast Concrete

- Precast concrete is manufactured in a factory and the transmitted to the site.
- The advantages are:
 - The pre-cast articles may be given the desired shape and finish with accuracy.
 - The labour required in the manufacturing process of pre-cast units can be easily trained.
 - Concrete of superior quality is produced
 - The pre-cast structures can be dismantled when required and they can then be suitably used elsewhere.
 - The work can be completed in short time.



Precast panels





IIT madras low cost building



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